

TU257 INFORMATION SYSTEMS

ASSIGNMENT 2 50%

CASE STUDY

In this case study, you are going to design a database for a particular scenario. The solution should provide:

- A. Conceptual design
 - a. Full list of requirements and assumptions
 - b. Rough ER Diagram (min 5 entities- max 10 entities)
 - c. Final key based ER Diagram
- B. Logical design
 - a. Final design in 3NF
 - b. Use a screenshot of the information required by the customer to make sure you are not forgetting any data.
 - c. Show any normalization steps or alternatively discuss how your design fulfills the 3NF.
- C. Physical Design
 - a. MySQL implementation
 - b. Indexing
 - c. Sample Queries

Choose your case study from:

- I. Airline booking system
- II. Maternity Hospital in-patient management

Submission

Deadline: 22th December 2024

- 1) A report including parts A, B and a description of the different components and features of C including rationale. This report should include all diagrams (do not submit your diagrams as separate files). A video report can be submitted instead of the written report as long as a document is used including all the figures and details.
- 2) Video of Physical design queries executed and explained.
- 3) Dump file.

Marking scheme

	Basic level	Added complexity/completeness
Conceptual Design description and assumptions	4	6
Conceptual design ER diagram including main entities and cardinality	2	4
Conceptual design Final ER diagram full list of attributes and PKs/FKs	4	4
Logical design		
Screenshots of typical data	2	2
Normalisation steps	3	6
3NF	2	2
Physical design		
Creation of schema	8	12
Sample basic queries	2	4
Sample complex queries	4	8
Optimisation	1	2

* Please note that custom marks in between levels can also be awarded.